



NEW PRODUCTS

COMPONENTS

DC/DC converter modules

Traco Power has announced the addition of TMDC 06 and TMDC 10 series of 6W and 10W products to its TMDC family of industrial DC/DC converter modules.



These modules are fully encapsulated with terminal strip connections and equipped with integrated filters that limit conducted and radiated emissions (EN 55032 class A) and increase the modules' EMC immunity (EN 55035) without the need for any external components.

Traco Power

www.tracopower.us

High-frequency IGBTs

STMicroelectronics' HB2 650V IGBT series delivers efficiency and performance gains for medium- and high-speed applications such as PFC converters, welders, uninterruptible power supplies and solar inverters, leveraging the company's Trench Field Stop technology. The series also includes automotive-eligible devices meeting AEC-Q101 Rev. D.

Joining the STPOWER portfolio, the new HB2 series has outstanding conduction performance thanks to low V_{CEsat} of 1.55V. At the same time, dynamic behaviour is enhanced due to reduced gate charge that enables fast switching at low gate current. Outstanding thermal performance helps maximise reliability and power density.

STMicroelectronics India

www.st.com

Schottky barrier diodes

Microchip's 700V and 1200V silicon carbide (SiC) Schottky barrier diodes (SBDs) join the existing portfolio of SiC power modules. The SBDs offer more efficient switching at higher frequencies and pass ruggedness tests at levels considered critical for guaranteeing long-term reliability. These perform approximately 20 per cent better than other SiC diodes in unclamped inductive switching (UIS) ruggedness tests that measure how well devices withstand degradation or premature failure under avalanche conditions, which occur when a voltage spike exceeds the device's breakdown voltage.

Microchip Technology Inc.

www.microchip.com

Microcontrollers

Renesas Electronics has introduced the RX72T group of 32-bit motor control microcontrollers (MCUs) with dedicated hardware accelerator IP to perform the complex, high-speed computations



required for motor control in robots and other industrial equipment. The RX72T group delivers superior performance, achieving an 1160 CoreMark score as measured by EEMBC Benchmarks—the highest level for a 5V MCU operating at 200MHz.

Renesas Electronics Singapore Pte Ltd

www.renesas.com

Gas sensor

Toshniwal Sensing Devices offers FIGARO, a Japanese carbon-dioxide gas sensor (model CDM7160). The module

uses a compact NDIR carbon-dioxide sensor, featuring excellent performance characteristics, including high accuracy and low power consumption. Two detector elements inside the module make absolute measurement possible. Every module is individually calibrated, and is provided with both a UART and I2C digital interface. The CDM7160 module can be used in a wide range of applications such as ventilation controls for the improvement of energy savings and to assure a good indoor climate.

Toshniwal Sensing Devices Pvt Ltd

www.tsdp.com

Digital attenuator

Analog Devices' ADRF5720 is a silicon, 6-bit digital attenuator with 31.5dB attenuation control range in 0.5dB steps. This device operates from 9kHz to 40GHz with better than 4.5dB insertion loss and excellent attenuation accuracy. ATTIN port of ADRF5720 has a radio frequency input power handling capability of 27dBm average and 30dBm peak for all states.

Analog Devices

www.analog.com

DC-DC converter

Artesyn Embedded Technologies has extended its range of low-power-output industrial DC-DC converters with the launch of ATA series 10-watt isolated DC-DC converters, comprising fourteen



variants. Targeting a diverse range of applications, including instrumentation, data communication, telecommunication and computer peripheral equipment, industrial automation and mobile battery-powered systems, the compact form factor of the ATA series minimises occupied board space and offers very high power density.

Artesyn Embedded Technologies

www.artesyn.com